## Unit 1 Progress Check: MCQ Part B

**1.** Consider the following code segment.

```
double a = 7;
int b = (int) (a / 2);
double c = (double) b / 2;
System.out.print(b);
System.out.print(" ");
System.out.print(c);
```

What is printed as a result of executing the code segment?

(A)

3 1.0

(B) ✓

3 1.5

(C)

3.5 1.5

(D)

3.5 1.75

**Answer B**

Correct. The value of a / 2 is 3.5. When cast to an int, the digits to the right of the decimal point are truncated, so b contains the value 3. The value c is calculated by dividing 3.0 by 2, resulting in 1.5.

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**2.** Assume that x is a double variable with a positive value. Which of the following code segments can be used to round x to the nearest integer and store the rounded value in the variable result?

(A)

```
int result = (int) x;
```

(B)

```
int result = (int) x + 0.5;
```

(C) ✓

```
int result = (int)(x + 0.5);
```

(D)

```
int result = (int) x + (int) 0.5;
```

### Answer C

Correct. This code segment adds 0.5 to x, then casts the result to an int, which removes the digits to the right of the decimal point. For example, if x is 4.2, the code segment adds 4.2 + 0.5, then casts the result 4.7 to the int 4. As another example, if x is 7.6, the code segment adds 7.6 + 0.5, then casts the result 8.1 to the int 8. In both of these examples, the original value of x is rounded to the nearest integer.

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**3.** A store sells rope only in whole-foot increments. Given three lengths of rope, in feet, the following code segment is intended to display the minimum length of rope, in feet, that must be purchased so that the rope is long enough to be cut into the three lengths.

For example, for lengths of 2.8 feet, 3 feet, and 5 feet, the minimum length of rope that must be purchased is 11 feet. For these values, the code segment should display 11. As another example, for lengths of 1.1 feet, 3.2 feet, and 2 feet, the minimum length of rope that must be purchased is 7 feet. For these values, the code segment should display 7.

In the following code segment, len1, len2, and len3 are properly declared and initialized double variables representing the three lengths of rope.

```
double total = len1 + len2 + len3;
int minLength = (int) (total + 0.5);
System.out.print(minLength);
```

Which of the following best describes the behavior of the code segment?

(A) The code segment works as intended for all nonnegative values of len1, len2, and len3.

(B) The code segment works as intended but only when the sum of the three lengths is an integer or the decimal part of the sum of the three lengths is greater than or equal to 0.5. ✓

(C) The code segment does not work as intended for some values of len1, len2, and len3 because these values must be cast to int before they are added together.

(D) The code segment does not work as intended for some values of len1, len2, and len3 because minLength should be declared as a double and then cast to an int in the print statement.

### Answer B

Correct. If the sum of the three lengths is an integer, then the exact amount of rope that is needed is returned. If the decimal part of the sum of the three lengths is greater than or equal to 0.5, then the code segment will correctly round the double value up to the nearest integer value. If the decimal part of the sum of the three lengths is less than 0.5, then the code segment will correctly round the double value down to the nearest integer value. However, to print the correct value, the code segment must always round up to the nearest integer to guarantee that the minimum length of rope is long enough to be cut into the three lengths.

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**4.** Consider the following code segment.

```
int x = 0;
x++;
x += 1;
x = x + 1;
x -= -1;
System.out.println(x);
```

What is printed when the code segment is executed?

(A)

1

(B)

2

(C)

3

(D) ✓

4

### Answer D

Correct. After x is initialized to 0, each of the next four statements increments x by 1, resulting in 4.

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**5.** Which of the following code segments can be used to print the output 2526?

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(A) ✓

```
int val = 5;

val *= val;

System.out.print(val);

val++;

System.out.print(val);
```

(B)

```
int val = 5;

val *= val;

System.out.print(val);

val *= 1;

System.out.print(val);
```

(C)

```
int val = 5;

val += val;

System.out.print(val);

val++;

System.out.print(val);
```

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(D)

```
int val = 5;

val += val;

System.out.print(val);

val *= 1;

System.out.print(val);
```

### Answer A

Correct. The statement val *= val; is equivalent to val = val * val;. It multiples val times itself and stores the result, which is 25, in val. The statement val++; is equivalent to val = val + 1;. It adds 1 to val and stores the result, 26, in val.

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**6.** Consider the following code segment.

```
int x = 0;              // line 1
x++;                    // line 2
System.out.print(x);    // line 3
x += 1;                 // line 4
System.out.print(x);    // line 5
x *= 2;                 // line 6
System.out.print(x);    // line 7
x *= 4;                 // line 8
System.out.print(x);    // line 9
```

The code segment is intended to produce the following output. The code segment does not work as intended.

1248

Which of the following changes can be made so that the code segment works as intended?

(A) Interchanging lines 2 and 4

(B) Interchanging lines 6 and 8

(C) Changing line 6 to `x += 2;`

(D) Changing line 8 to `x *= 2;` ✓

### Answer D

Correct. In the given code segment, the initial value of `x` is 0. The expressions `x++` adds 1 to `x`, causing 1 to be printed. The expression `x += 1` adds 1 to `x`, causing 2 to be printed. The expression `x *= 2` multiplies `x` by 2, causing 4 to be printed. The expression `x *= 4` multiples `x` by 4, causing 16 to be printed. By changing line 8 to `x *= 2;`, the value of `x` (4 at this point) is multiplied by 2, causing 8 to be printed.

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**7.** The following is an excerpt of a class specification that appears in an API library.

public class Vehicle

The Vehicle class is used to create objects that represent vehicles.

The Vehicle class has make, model, and year variables that hold information about the vehicle's characteristics.

The Vehicle constructor initializes a Vehicle object with the given make, model, and year.

The getMake(), getModel(), and getYear() methods return information about the vehicle.

Based on the class specification, which of the following descriptions is accurate?

(A) Vehicle objects are attributes of the variables make, model, and year.

(B) Vehicle objects are behaviors of the variables make, model, and year.

(C) The variables make, model, and year are attributes of Vehicle objects. ✓

(D) The variables make, model, and year are behaviors of Vehicle objects.

**Answer C**

Correct. Attributes hold information related to a class and are stored in variables.

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**8.** The following is an excerpt of a class specification that appears in an API library.

public class Robot

A Robot specifies a robot that moves around a two-dimensional coordinate grid.

The Robot class has xCoordinate and yCoordinate variables that hold information about the robot's location on the grid.

The Robot constructor initializes a Robot object with the given coordinates.

The move() and rotate() methods are used to modify the robot's location on the grid.

Based on the class specification, which of the following descriptions is accurate?

(A) Robot objects are attributes of the methods move and rotate.

(B) Robot objects are behaviors of the methods move and rotate.

(C) The methods move and rotate represent attributes of Robot objects.

(D) The methods move and rotate represent behaviors of Robot objects. ✓

**Answer D**

Correct. Behaviors describe what instances of a class can do and are defined by methods.

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**9.** The following is an excerpt of a class specification that appears in an API library.

public class Parcel

A Parcel class is used to create objects that represent the status of packages that are delivered to customers.

The Parcel class has status and location variables that hold information about a package's shipping status and location.

The Parcel constructor initializes a Parcel object with its initial status and location.

The updateStatus() and updateLocation() methods are used to update the status and location of a package.

Based on the class specification, which of the following is a true statement about Parcel objects?

(A) status and location are attributes; updateStatus and updateLocation are behaviors. ✓

(B) status and location are behaviors; updateStatus and updateLocation are attributes.

(C) status and updateStatus are attributes; location and updateLocation are behaviors.

(D) status and updateStatus are behaviors; location and updateLocation are attributes.

**Answer A**

Correct. Attributes hold information related to a class and are stored in variables. Behaviors describe what instances of a class can do and are defined by methods.

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**10.** The following code segment appears in a method. In the code segment, num1 and num2 are int variables. The method is intended to print the sum of num1 and num2.

```
int result = num1 + num2;
System.out.println(result);
```

Which of the following preconditions for the method is most appropriate to avoid an overflow error?

(A)

```
/** Precondition: num1 and num2 are both positive. */
```

(B)

```
/** Precondition: num1 is not equal to num2 */
```

(C)

```
/** Preconditions: num1 is between Integer.MIN_VALUE
 *                 and Integer.MAX_VALUE, inclusive.
 *                 num2 is between Integer.MIN_VALUE
 *                 and Integer.MAX_VALUE, inclusive.
 */
```

(D) ✓

```
/** Precondition: (num1 + num2) is between Integer.MIN_VALUE
 *                and Integer.MAX_VALUE, inclusive.
 */
```

### Answer D

Correct. An overflow error occurs when an expression evaluates to an expression outside the allowable range, which is less than Integer.MIN_VALUE or greater than Integer.MAX_VALUE. An overflow error can be avoided by specifying that the sum must be between these two values.

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**11.** The following code segment appears in a method. In the code segment, y is an int variable.

```
int x = 5 / y;
System.out.println(x);
```

Which of the following preconditions for the method is most appropriate to avoid an ArithmeticException?

(A)

```
/** Precondition: y is equal to 0. */
```

(B)

```
/** Precondition: y is less than 5. */
```

(C) ✓

```
/** Precondition: y is not equal to 0. */
```

(D)

```
/** Precondition: y is not negative. */
```

**Answer C**

Correct. An ArithmeticException will occur when an integer is divided by 0. Specifying that y is not equal to 0 will prevent this error from occurring.

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**12.** The following code segment appears in a method. In the code segment, x and y are double variables.

```
double result = (x - y) / 2.0;
System.out.println(result);
```

Which of the following preconditions for the method is most appropriate to ensure that the value printed by the code segment is always positive?

(A)

```
/** Precondition: x and y are both positive. */
```

(B)

```
/** Precondition: x and y are both even. */
```

(C) ✓

```
/** Precondition: x > y */
```

(D)

```
/** Precondition: x < y */
```

**Answer C**

Correct. If x > y, then the difference x - y will be positive, and this difference divided by 2.0 will also be positive.

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**13.** Consider the following descriptions of two methods that appear in the same class.

| Method Signature | Explanation |
|---|---|
| public void methodA(int arg) | Calls methodB with the value of arg * 10 |
| public void methodB(int arg) | Displays the value of arg + 10 |

Consider the call methodA(4), which appears in a method in the same class. What, if anything, is printed as a result of the call methodA(4)?

(A)
40

(B) ✓
50

(C)
140

(D) Nothing is printed.

**Answer B**

Correct. The value 4 is passed to methodA and stored in methodA's parameter arg. This value is multiplied by 10. The result 40 is then passed to methodB and stored in methodB's parameter arg. The value 10 is added to arg, and the result 50 is printed.

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**14.** Consider the following description of method printValues.

| Method Signature | Explanation |
|---|---|
| public void printValues(int numTimes, int val) | Prints the value of val a total of numTimes times, then moves the cursor to a new line. |

Consider the following code segment, which appears in the same class as printValues.

```
printValues(2, 3);
printValues(4, 5);
```

What is printed as a result of executing the code segment?

(A)

```
222

44444
```

(B)

```
2222

33333
```

(C) ✓

```
33

5555
```

(D)

```
44

555
```

**Answer C**

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Correct. The first call to printValues passes 2 to the parameter numTimes and 3 to the parameter val. This call results in the output 33. The second call to printValues passes 4 to the parameter numTimes and 5 to the parameter val. This call results in the output 5555.

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**15.** Consider the following description of method printSomething.

| Method Signature | Explanation |
|---|---|
| public void printSomething(int num, boolean val) | Prints the value of val immediately followed by the value of num - 1. |

Consider the following code segment, which appears in the same class as printSomething.

```
printSomething(1, true);
printSomething(2, true);
```

What is printed as a result of executing the code segment?

(A)

0true1true

(B)

1true2true

(C) ✓

true0true1

(D)

true1true2

### Answer C

Correct. The first call to printSomething passes 1 to the parameter num and true to the parameter val. The value of num is decreased by 1, then val and num are each printed, resulting in the output true0. The second call to printSomething passes 2 to the parameter num and true to the parameter val. The value of num is decreased by 1, then val and num are each printed, resulting in the output true1.